

"Find the right candidate. Not just the right keywords."

1. THE BIG IDEA IN 30 SECONDS

The Core Innovation: Automated resume filtering is fundamentally broken between *rigid regex scanners* that reject capable students over synonym variations, and *blind LLM prompts* that invent unexplainable scores. **InternLoom AI** introduces a transparent, explainable **Dual-Engine Architecture**: pairing **local vector semantic embeddings** (BAAI/bge-small-en-v1.5) with **multi-tier lexical term saturation (BM25 + TF-IDF)**, wrapped in an Apple-inspired recruitment interface featuring automated HR bias scanning.

2. PROBLEM VS. SOLUTION BREAKDOWN

The Problem: Broken Campus Screening	The Solution: InternLoom AI Dual-Engine
Vocabulary Mismatch: Students describe skills variably (e.g. <i>FastAPI</i> vs <i>REST APIs</i>, <i>Node</i> vs <i>Node.js</i>); traditional regex rates them as 0% match. LLM Hallucinations: Asking ChatGPT/Claude to "score out of 100" creates non-reproducible, unexplainable results with privacy exposure. Institutional JD Elitism: JDs require "Tier-1 only" or "3+ yrs experience" for internships, unnecessarily shrinking qualified talent pools by 80%+. Messy Submissions: Formatting typos (<i>"reaktjs"</i>) and non-standard dates cause parsers to discard viable candidates.	Local Semantic AI: Computes true conceptual meaning and domain depth using local BGE embeddings and scikit-learn cosine similarity. Deterministic Mathematical Fusion: Fuses BM25 token relevance with semantic cosine distance with zero blind LLM guesswork. Automated Bias Scanner: Audits JDs for exclusionary institutional criteria and provides 1-click inclusive rephrasing. Resilient Ingestion: Levenshtein typo tolerance and section inference gracefully normalize student resumes without scoring penalties.

3. MATHEMATICAL SCORING FORMULATION

$$\text{Score}(\text{Resume}, \text{JD}) = [w_{\text{keyword}} \times \text{Score}_{\text{BM25}} / \text{Ontology}] + [w_{\text{semantic}} \times \text{Score}_{\text{BGE-Cosine}}]$$

Placement officers have complete transparency through an interactive weight slider (10% to 90%), balancing strict tool compliance against conceptual domain breadth.

4. UNIQUE VALUE PROPOSITIONS & INNOVATIONS

- 1. 100% Deterministic • Zero Blind LLMs:** All embeddings run locally; scores are mathematical, reproducible, and explainable down to 2 decimal places.

3. Resilient Messy Resume Ingestion: Recovers from spelling slips (*"reaktjs"* → *React*), informal dates (*"Summer '24"*), and missing headers.
- 2. Head-to-Head Candidate Comparator:** Direct side-by-side modal displaying exact score deltas, sub-scores, and unique skill advantages.

4. Grounded AI Recruiter Q&A: Natural-language query drawer providing verifiable answers based strictly on resume textual citations.

5. HACKATHON JUDGING MATRIX COMPLIANCE

Evaluation Criteria	Weight & Target	InternLoom AI Implementation & Evidence
Semantic & Keyword Fusion	35% WEIGHT	Dual-engine combining BM25Okapi, TF-IDF n-grams, and local BAAI/bge-small-en-v1.5 cosine embeddings.
Quality of Ranking Spread	20% WEIGHT	Evaluated across 24 realistic campus candidates with distinct full-stack, frontend, backend, and gap profiles.
Explainable Justifications	20% WEIGHT	Automated evidence snippets, domain sub-scores, and explicit top-3 candidate hiring arguments.
Working End-to-End Demo	15% WEIGHT	Full-stack production setup: Python FastAPI (Port 8000) + React 19 / Express / Vite (Port 3000).
Innovation & Bonus Features	10% WEIGHT	Automated JD Bias Scanner, Resilient Messy Ingestion, and Grounded AI Recruiter natural-language chat.